

Given: Due:	Further Maths (Core Pure) HW 11	Name:
Chapter 7: Matrix transformation		
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.		
91% (A*)	76% (A)	66% (B)
56% (C)	50% (D)	40% (E)
<i>Useful resources:</i> Your notes from recent lessons & your text book <u>Although most matrix arithmetic can be done on your calculator, show all steps to reach the answer and do check results on your calculator</u>		
Questions - do these on lined paper		
1) $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}$ (a) Find $\mathbf{A}^{-1}$ (2) The transformation represented by the matrix $\mathbf{B}$ followed by the transformation represented by the matrix $\mathbf{A}$ is equivalent to the transformation represented by the matrix $\mathbf{P}$. (b) Find $\mathbf{B}$, giving your answer in its simplest form. (3)		
2) (i) In each of the following cases, find a 2×2 matrix that represents (a) a reflection in the line $y = -x$, (b) a rotation of 135° anticlockwise about $(0, 0)$, (c) a reflection in the line $y = -x$ followed by a rotation of 135° anticlockwise about $(0, 0)$. (4) (ii) The triangle T has vertices at the points $(1, k)$, $(3, 0)$ and $(11, 0)$, where k is a constant. Triangle T is transformed onto the triangle T' by the matrix $\begin{pmatrix} 6 & -2 \\ 1 & 2 \end{pmatrix}$ Given that the area of triangle T' is 364 square units, find the value of k. (6)		
3) $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ The transformation represented by $\mathbf{B}$ followed by the transformation represented by $\mathbf{A}$ is equivalent to the transformation represented by $\mathbf{P}$. (a) Find the matrix $\mathbf{P}$. (2)		

Triangle T is transformed to the triangle T' by the transformation represented by $\mathbf{P}$.

Given that the area of triangle T' is 24 square units,

(b) find the area of triangle T .

(3)

Triangle T' is transformed to the original triangle T by the matrix represented by $\mathbf{Q}$.

(c) Find the matrix $\mathbf{Q}$.

(2)

4) The transformation U , represented by the 2×2 matrix $\mathbf{P}$, is a rotation through 90° anticlockwise about the origin.

(a) Write down the matrix $\mathbf{P}$.

(1)

The transformation V , represented by the 2×2 matrix $\mathbf{Q}$, is a reflection in the line $y = -x$.

(b) Write down the matrix $\mathbf{Q}$.

(1)

Given that U followed by V is transformation T , which is represented by the matrix $\mathbf{R}$,

(c) express $\mathbf{R}$ in terms of $\mathbf{P}$ and $\mathbf{Q}$,

(1)

(d) find the matrix $\mathbf{R}$,

(2)

(e) give a full geometrical description of T as a single transformation.

(2)

5)

$$\mathbf{A} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

(a) Describe fully the single geometrical transformation U represented by the matrix $\mathbf{A}$.

(3)

The transformation V , represented by the 2×2 matrix $\mathbf{B}$, is a reflection in the line $y = -x$.

(b) Write down the matrix $\mathbf{B}$.

(1)

Given that U followed by V is the transformation T , which is represented by the matrix $\mathbf{C}$,

(c) find the matrix $\mathbf{C}$.

(2)

(d) Show that there is a real number k for which the point $(1, k)$ is invariant under T .

(4)

6)

$$\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$



- (a) Describe fully the single geometrical transformation U represented by the matrix $\mathbf{P}$. (2)
- The transformation U maps the point A , with coordinates (p, q) , onto the point B , with coordinates $(6\sqrt{2}, 3\sqrt{2})$.
- (b) Find the value of p and the value of q . (3)
- The transformation V , represented by the 2×2 matrix $\mathbf{Q}$, is a reflection in the line with equation $y = x$.
- (c) Write down the matrix $\mathbf{Q}$. (1)
- The transformation U followed by the transformation V is the transformation T . The transformation T is represented by the matrix $\mathbf{R}$.
- (d) Find the matrix $\mathbf{R}$. (3)
- (e) Deduce that the transformation T is self-inverse. (1)

END OF HOMEWORK

Total = 49 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.